

Machine Learning Basics

Machine learning is a subset of artificial intelligence that enables systems to learn and improve from experience without being explicitly programmed.

Supervised Learning: The algorithm learns from labeled training data, mapping inputs to outputs. Common algorithms include linear regression, decision trees, and support vector machines.

Unsupervised Learning: The algorithm finds patterns in unlabeled data. Common techniques include clustering (K-means, hierarchical) and dimensionality reduction (PCA, t-SNE).

Reinforcement Learning: An agent learns by interacting with an environment, receiving rewards or penalties. It is used in robotics, gaming, and autonomous systems.

Neural Networks: Inspired by the human brain, neural networks consist of layers of interconnected nodes. Deep learning uses many layers to model complex patterns.

Key concepts include training/testing splits, overfitting, regularization, cross-validation, and evaluation metrics like accuracy, precision, recall, and F1-score.